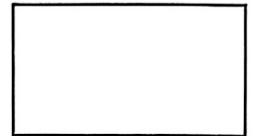




East West University
Department of Mathematics and
Data Science
MAT 205
Section: 8



Marks Obtained

Name:

ID. No.:

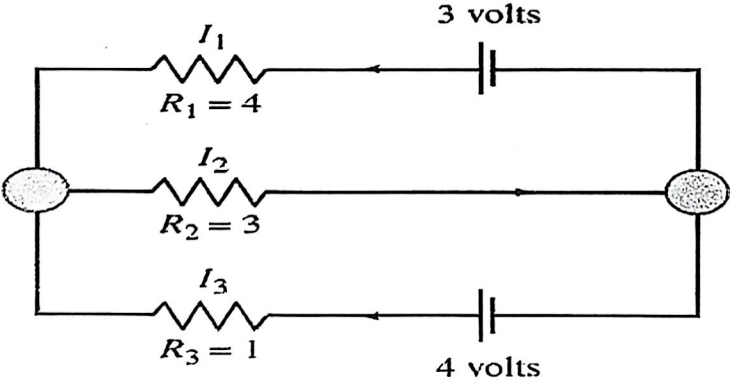
Quiz No.: 02

Date: 01.07.2026

Topics: REF, Rank, System of Equations, Applications

Total Marks: 20

Time: 30 minutes

1.	Compare Cramer's Rule, the Matrix (Inverse) Method, Gaussian Elimination, and Gauss-Jordan Elimination as techniques for solving a system of linear equations. State one advantage and one limitation of each method.	[4]
2.	Find the general solution of the following homogeneous system of linear equations using Gauss-Jordan elimination method. Also, determine two particular solutions of this system. $x + y + z + w = 0$ $2x + 2y + 2z + 2w = 0$ $x - y + z - w = 0$ $3x + y + 3z + w = 0$	[10]
3.	Form a system of linear equations to determine the currents I_1 , I_2 and I_3 for the electrical network shown in the following Figure. 	[6]